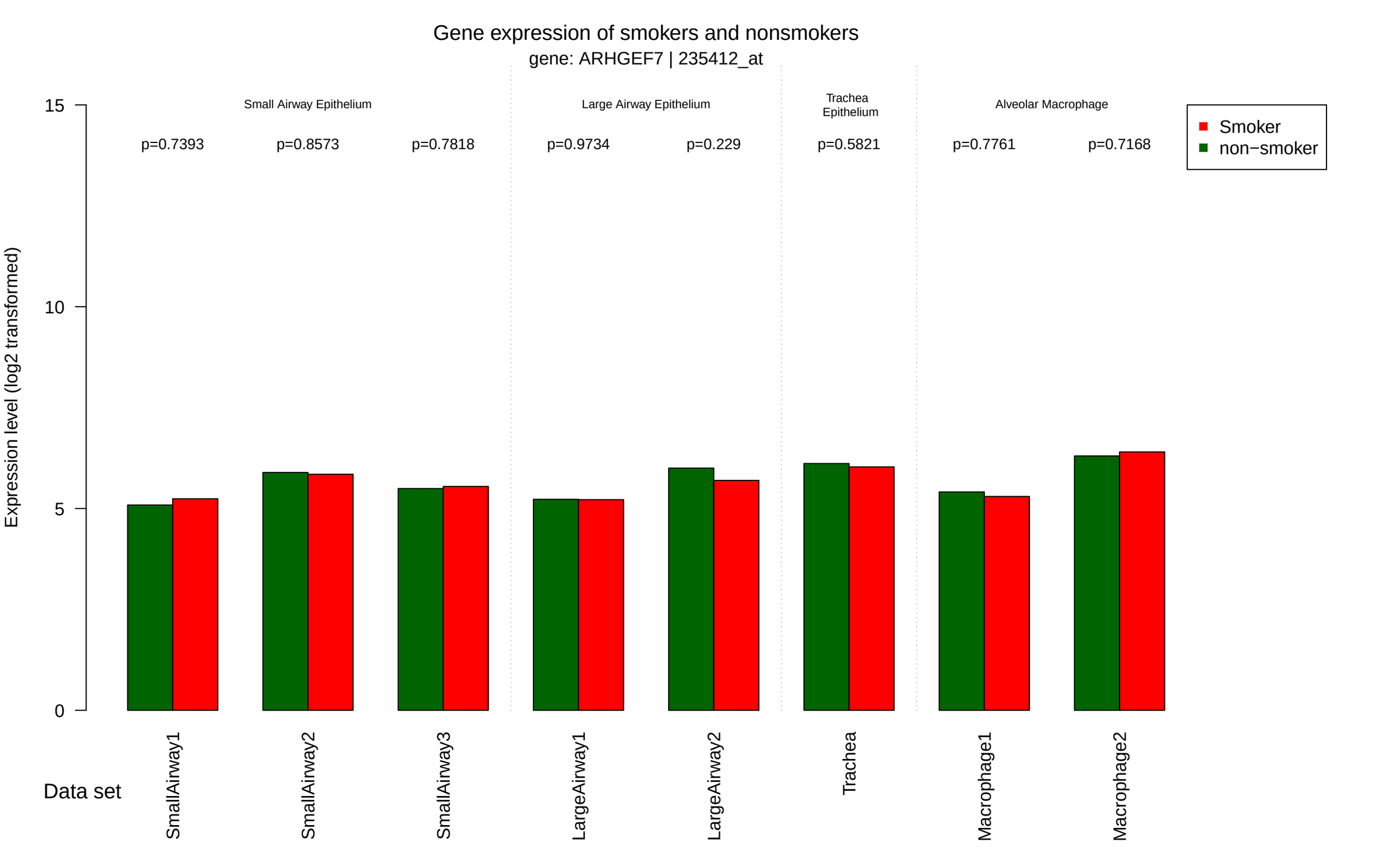
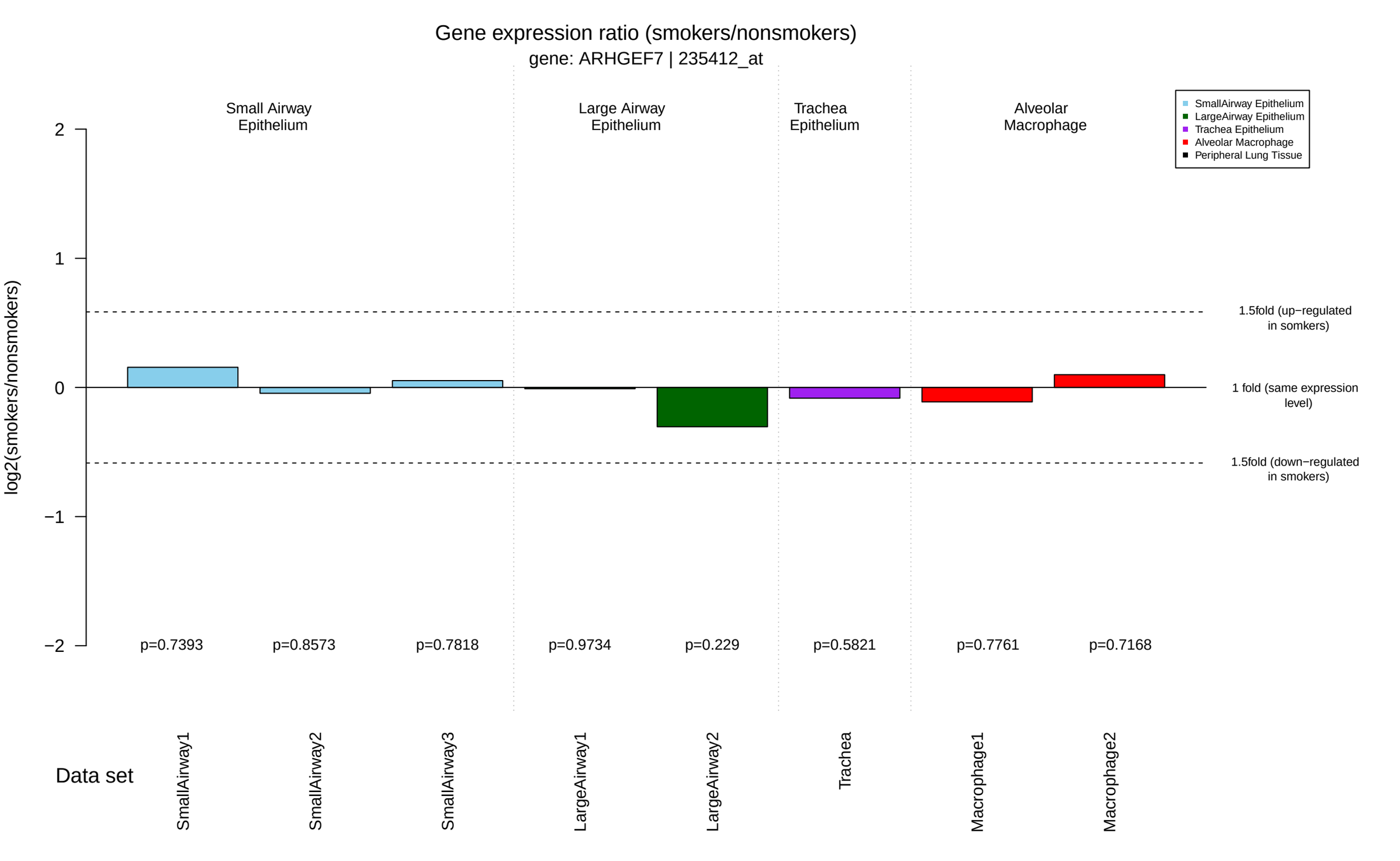
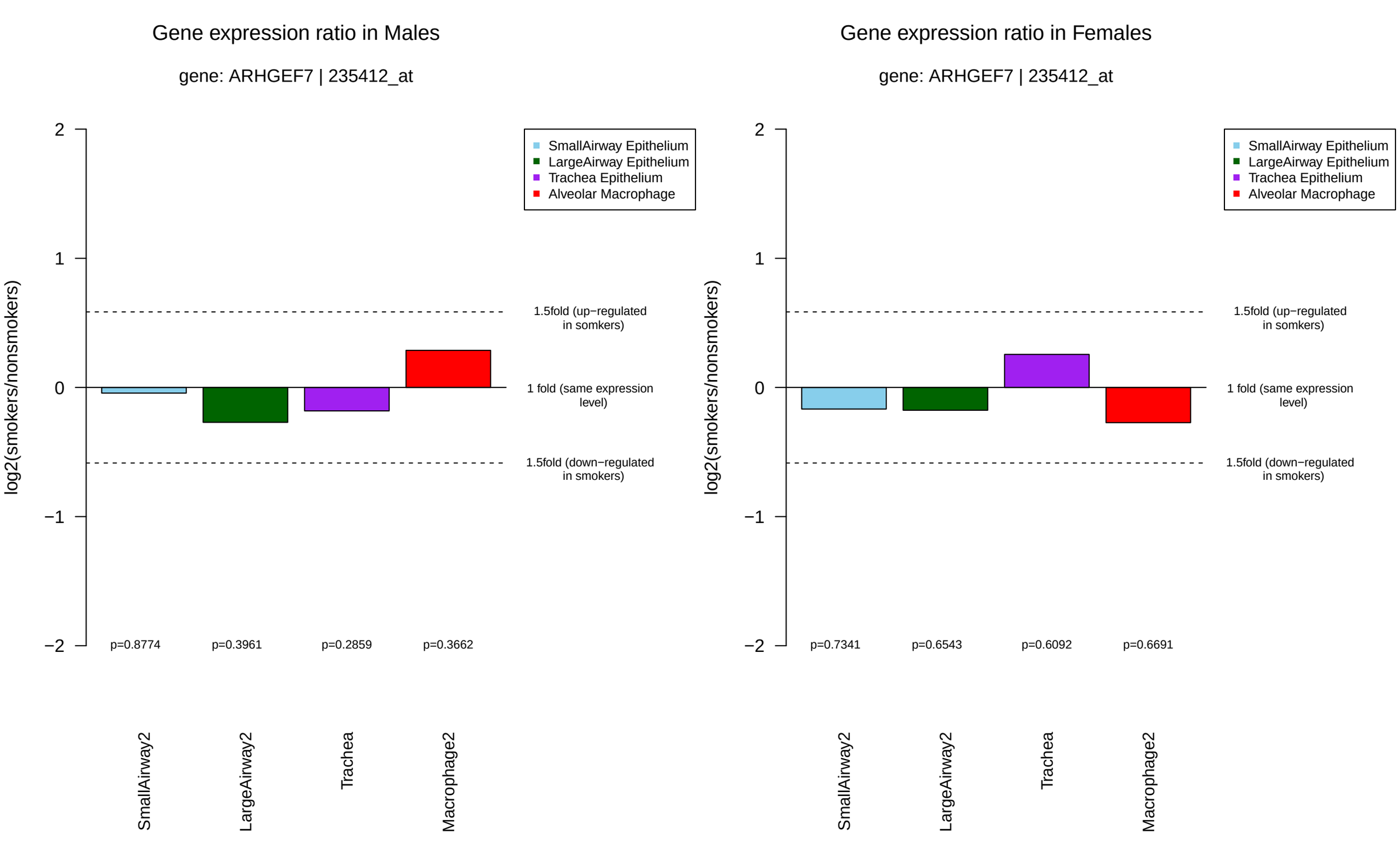
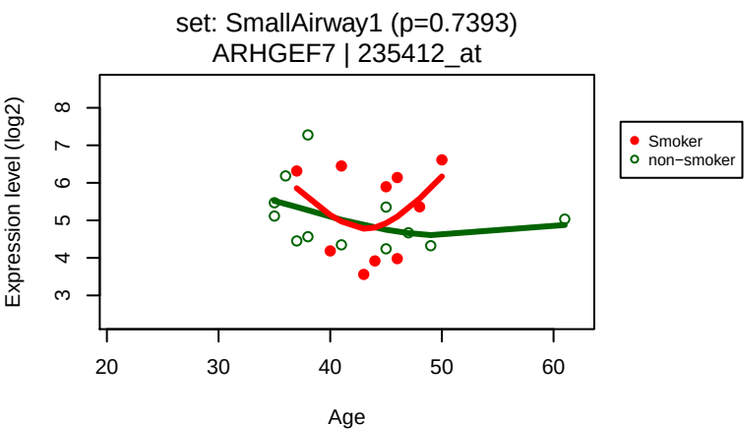
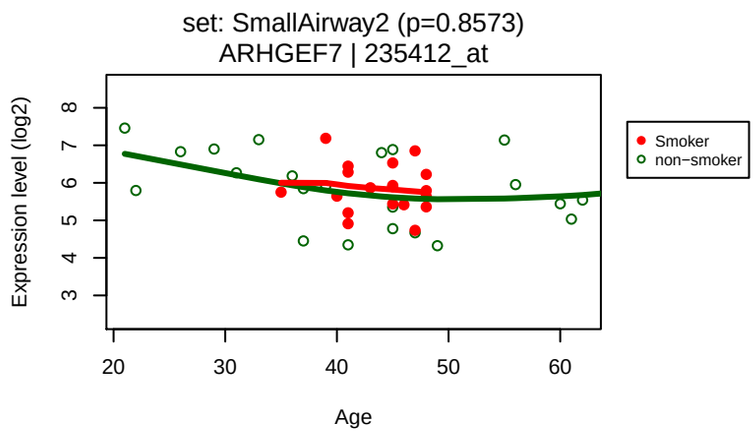
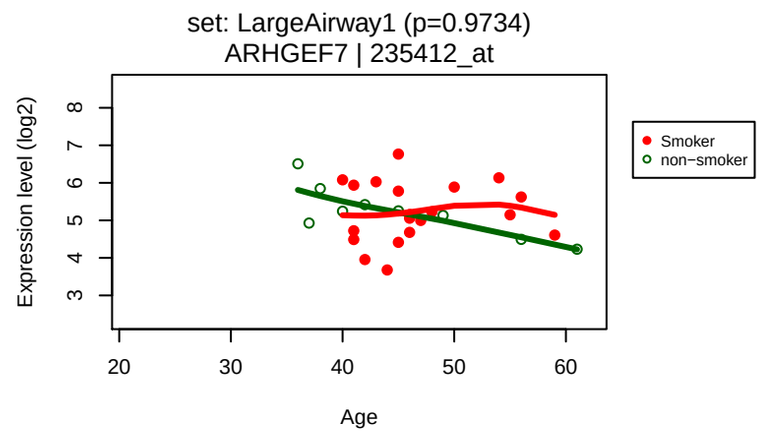

Gene expression of smokers and nonsmokers
gene: ARHGEF7 | 235412_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7393
p=0.8573
p=0.7818
p=0.9734
p=0.229
p=0.5821
p=0.7761
p=0.7168
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

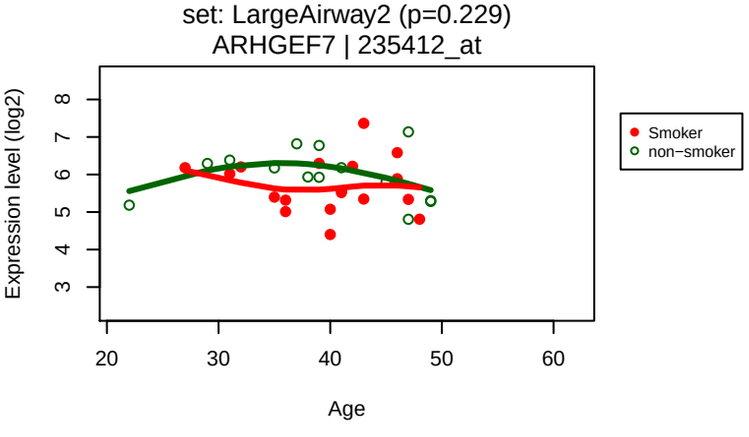

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF7 | 235412_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7393
p=0.8573
p=0.7818
p=0.9734
p=0.229
p=0.5821
p=0.7761
p=0.7168
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

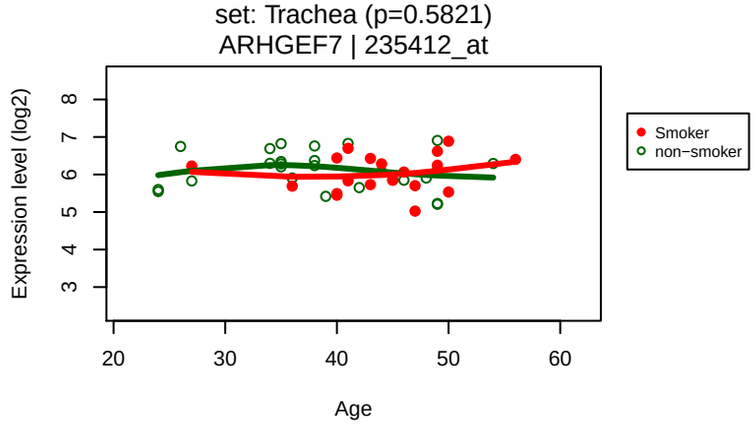

Gene expression ratio in Males
gene: ARHGEF7 | 235412_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8774
p=0.3961
p=0.2859
p=0.3662
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF7 | 235412_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7341
p=0.6543
p=0.6092
p=0.6691
SmallAirway2
LargeAirway2
Trachea
Macrophage2

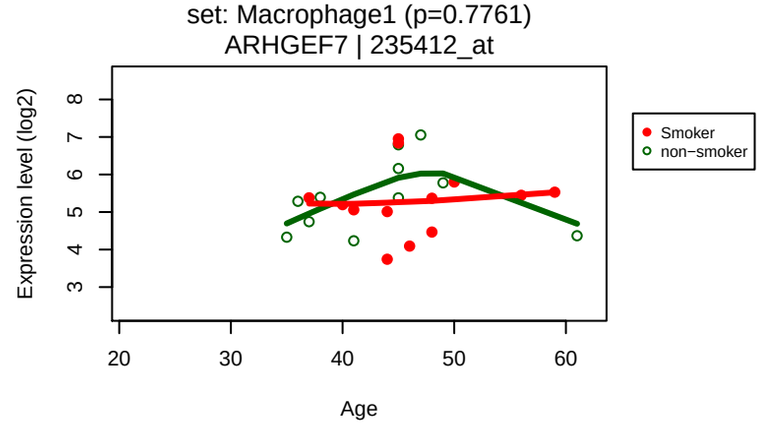

set: SmallAirway1 (p=0.7393)
ARHGEF7 | 235412_at
Expression level (log2)
Age
Smoker
non−smoker

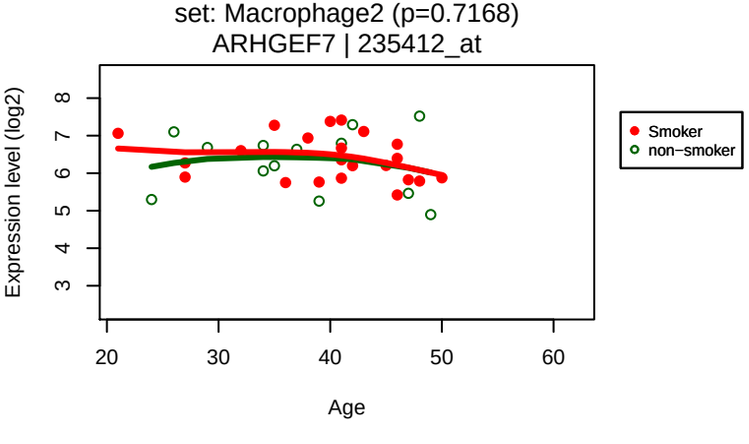

set: SmallAirway2 (p=0.8573)
ARHGEF7 | 235412_at
Expression level (log2)
Age
Smoker
non−smoker

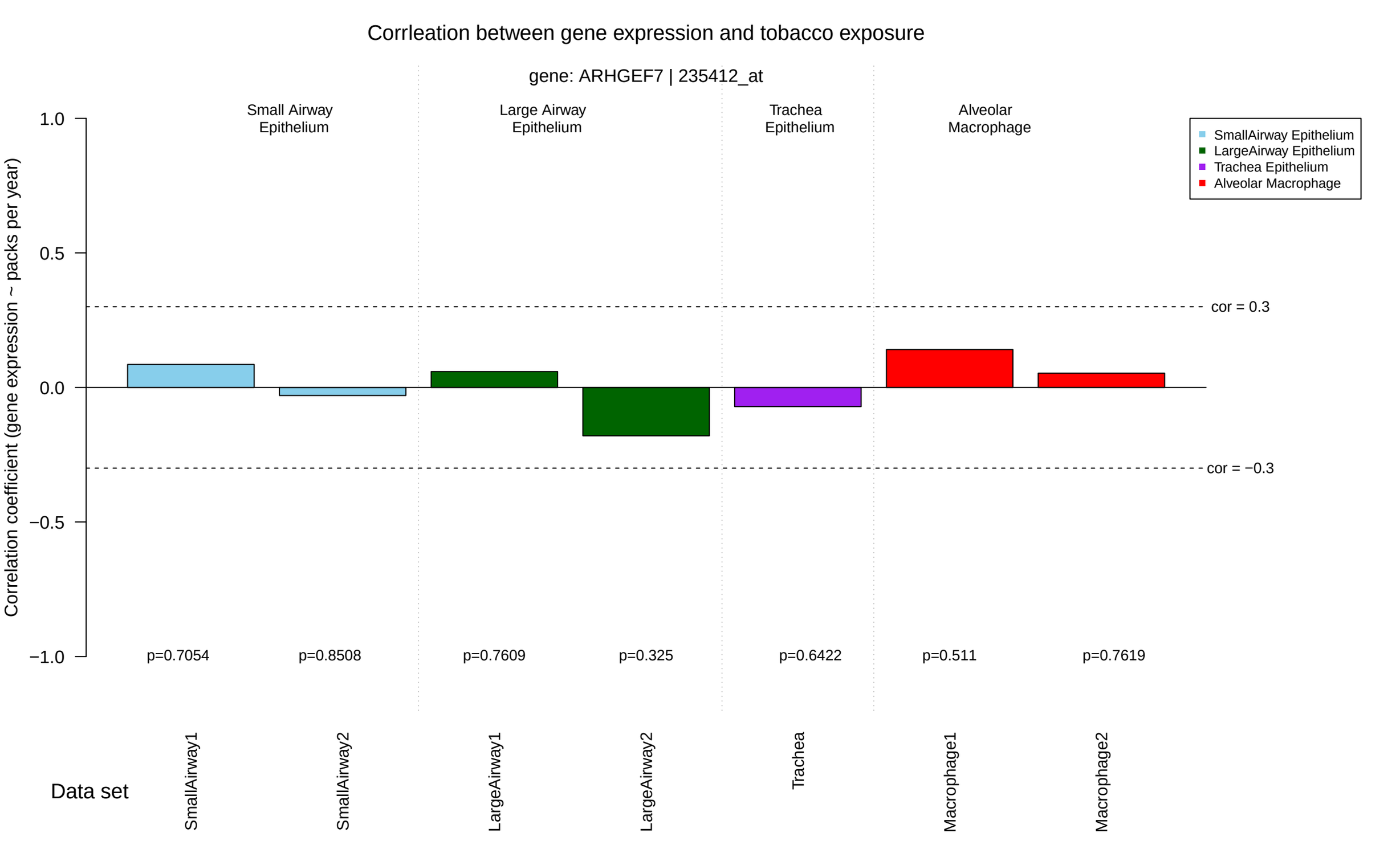

set: LargeAirway1 (p=0.9734)
ARHGEF7 | 235412_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.229)
ARHGEF7 | 235412_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5821)
ARHGEF7 | 235412_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7761)
ARHGEF7 | 235412_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7168)
ARHGEF7 | 235412_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF7 | 235412_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7054
p=0.8508
p=0.7609
p=0.325
p=0.6422
p=0.511
p=0.7619
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2